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Inventor(s)

SPINAZZOLA; Robert Stephen et al.

Hybrid Dry Adiabatic Cooling Chilled Water Plant for Data Centers and Other IT Environments

Abstract

A hybrid adiabatic cooling system for a data center includes an outer conduit loop including a feed segment and a return segment and an inner conduit loop fluidly coupled to the feed and return segments of the outer conduit loop. The hybrid adiabatic cooling system includes a hybrid adiabatic cooler fluidly coupled to the outer conduit loop to cool a first cooling liquid circulating from the return segment to the feed segment of the outer conduit loop. The hybrid adiabatic cooling system includes a direct liquid cooling unit fluidly coupled to the outer conduit loop and an air handling unit fluidly coupled to the inner conduit loop. The hybrid adiabatic cooling system operates in a free cooling mode when the outdoor air temperature is less than a first threshold temperature and a chiller mode when the outdoor air temperature is greater than the first threshold temperature.

Inventors: SPINAZZOLA; Robert Stephen (Lewes, DE), PENROD; Justin Neal (Fairfax, VA)

Applicant: Shumate Engineering, PLLC (Tysons, VA)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS AND INCORPORATION BY REFERENCE [0001] This application claims priority to U.S. Provisional Patent Application No. 63/561,437 filed on Mar. 5, 2024, which is incorporated by reference herein in its entirety for all purposes.

FIELD

[0002] The present disclosure relates to methods and systems for cooling a data center, particularly for using free cooling and chiller assisted cooling to maintain operation temperatures in a data center.

BACKGROUND

[0003] Data centers and other IT environments typically consist of a facility used to house computer systems and associated components, such as servers, telecommunications, and storage systems. Data centers generally include redundant or backup power supplies, redundant data communications connections, redundant environmental controls (e.g., air conditioning, fire suppression) and security devices. Large data centers operating at industrial scale operations consume as much electricity as a small city.

[0004] One of the main concerns of this industry is business continuity. Companies rely on their information systems to run their operations. If a system becomes unavailable, business operations may be impaired or stopped completely. Reliable infrastructure is necessary for IT operations to minimize any chance of disruption. Information security is also a concern, and for this reason, a data center usually has to offer a secure environment to minimize the chances of a security breach. A data center must therefore keep high standards for assuring the integrity and functionality of its hosted computer environment. This is accomplished through redundancy of both fiber optic cables and power, which includes emergency backup power generation.

[0005] The physical environment of a data center is rigorously controlled to maintain IT operations. Air conditioning is commonly used to control the temperature and humidity in the data center. For example, current ASHRAE's "Thermal Guidelines for Data Processing Environments" recommends a temperature in a range from 18° C. to 27° C. (64.4° F. to 80.6° F.) and a dew point in a range from -9° C. to 15° C. (15.8° F. to 59.0° F.) to maintain optimal data center conditions. The temperature in a data center will naturally rise due to the electrical power dissipating heat into the air as a byproduct of the computing process. Unless the heat is removed, the ambient temperature will rise, resulting in electronic equipment malfunction. By controlling the air temperature, the server components at the board level are kept within the manufacturer's specified temperature/humidity range. Air conditioning systems help control humidity by cooling the return space air to below the dew point. Too much humidity may lead to condensation on internal components. In case of a dry atmosphere with low humidity, ancillary humidification systems may add water vapor. But the added water vapor may cause static electric discharge problems, potentially damaging the data center's components.

[0006] Data center heat removal is one of the most essential of all critical IT environment processes. As computing equipment is becoming smaller and increasing in power density, the high concentration of computing equipment in data centers consumes more electricity than the previous generations of IT equipment, thereby generating more heat in data centers.

[0007] Power draw for data centers ranges from a few kilowatts (kW) for a rack of servers in a closet to several tens of megawatts (MW) for large facilities. For higher power density facilities,

electricity costs are a dominant operating expense. Precision cooling and heat rejection equipment is used to collect and transport unwanted heat energy in the data center to the outside atmosphere. [0008] A cooling architecture for a data center is fundamentally described by: (i) a particular heat removal method, (ii) a particular air distribution type, and (iii) the location of the cooling unit(s) that directly supplies cool air and chilled water to the IT equipment.

[0009] IT environments of data centers are cooled by transferring unwanted heat energy from the data center's IT equipment to the outdoors using fundamental heat methods as explained below.

[0010] Heat removal is the process of “moving” heat energy from the IT space of the data center to the outdoors. This can be accomplished as easily as using an air duct to “transport” heat energy to the cooling system located outdoors. However, “movement” of heat energy is generally accomplished by using a heat exchanger to transfer heat energy from one fluid to another (e.g. from air to water). A computer room air handler unit(s) (also called a CRAH or CRAHU), combined with a chiller(s), a pump(s), is known as a chilled water system.

[0011] Conventional CRAHUs are similar to computer room air conditioners (CRAC) in appearance, but work differently. In a chilled water system, the components of the refrigeration cycle are relocated from the computer room air conditioning systems to a device called a water chiller. The function of the water chiller, or chiller, is to produce chilled water. Conventional CRAHUs cool air (remove heat) by drawing warm air from the computer room through the CRAHU to transfer the energy to the chilled water by running it through the chiller. The chiller removes the heat from the warmer chilled water and transfers it to another stream of circulating water called condenser water, which flows through a device known as a cooling tower. The cooling tower rejects heat taken from the IT room to the outdoor environment by spraying warm condenser water at the top of the tower onto corrugated material (called fill). The water spreads out, and some of the water evaporates away as it drips and flows to the bottom of the cooling tower. A fan is commonly used to help promote evaporation by drawing air through the fill material. In the same manner as the human body is cooled by the evaporation of sweat, the small amount of water that evaporates from the cooling tower serves to lower the temperature of the remaining water. The chilled water is pumped in pipes from the chiller to the CRAHs located in the IT environment.

[0012] There are three main types of chillers distinguished by their use of water or air to dissipate heat from the returning chilled water. First, water-cooled chillers use a condenser water loop to dissipate the heat from the returning chilled water to the outside atmosphere. The heated condenser water is cooled by a cooling tower—the final step in dissipating the heat to the outdoors. Water-cooled chillers are typically located indoors. Second, water/glycol-cooled chillers, which look identical to water-cooled chillers, typically use a coolant that contains glycol at 30% concentration by volume. In water/glycol-cooled chillers, heat from the returning chilled water is transferred to a glycol loop, which ultimately dissipates the transferred heat to the outside atmosphere. The glycol flows via pipes to an outdoor-mounted device called a dry cooler (also known as a fluid cooler). The heat in the warm glycol-filled coil of the dry cooler is dissipated to the outside atmosphere by a fan drawing outdoor air over the dry cooler's coils. Water is sprayed on the glycol-filled coils, and the evaporation of water adiabatically cools the air drawn over the coil. Glycol-cooled chillers are typically located indoors. Third, air-cooled chillers use an air-cooled condenser integrated with a chiller to dissipate heat from the returning chilled water. Compressors are integrated with an air-cooled chiller and cool the closed loop chilled water. The chilled water typically contains glycol at approximately 30% by volume. This type of chiller is known as a packaged chiller and can be integrated into a cooling system. Air-cooled chillers are typically located outdoors.

BRIEF SUMMARY

[0013] The present disclosure provides a hybrid adiabatic cooling system having one continuous first cooling liquid (e.g., water) loop that provides two distinct fluid temperatures using the same system infrastructure. The hybrid adiabatic cooling system is configured to accommodate a variety of different ambient outdoor conditions, which impact the process fluid temperatures within the

loop. During low ambient temperature conditions, the hybrid adiabatic cooling system supplies the first cooling liquid at a desired operating temperature (e.g., 23° C., 73° F.) to both an air handling unit and a direct liquid cooling unit to address different heat loads of the data center without the use of chillers. As the ambient outdoor temperature exceeds a first temperature threshold (e.g., 18.3° C., 65.0° F.), a secondary pump and a chiller of the cooling system are initialized to maintain the first cooling liquid supplied to the air handling unit at the desired operating temperature (e.g., 22.8° C., 73.0° F.), while supplying the first cooling liquid to the direct liquid cooling unit at a maximum operating temperature (e.g., 32.2° C., 90.0° F.). All of these conditions are maintained using a single continuous piping network.

[0014] In some embodiments, the present disclosure provides a hybrid adiabatic cooling system. In some embodiments, the hybrid adiabatic cooling system includes an outer conduit loop that includes a feed segment to direct a first cooling liquid from outside the data center to inside the data center and a return segment to direct the first cooling liquid from inside the data center to outside the data center. In some embodiments, the hybrid adiabatic cooling system includes a hybrid adiabatic cooler disposed outside of the data center. The hybrid adiabatic cooler includes an outlet fluidly coupled to the feed segment of the outer conduit loop and an inlet fluidly coupled to the return segment of the outer conduit loop. The hybrid adiabatic cooler is configured to draw outdoor ambient air to cool the first cooling liquid circulating from the inlet of the hybrid adiabatic cooler to the outlet of the hybrid adiabatic cooler. In some embodiments, the hybrid adiabatic cooling system includes a direct liquid cooling unit disposed inside the data center, the direct liquid cooling unit including an inlet fluidly coupled to the feed segment of the outer conduit loop to receive the first cooling liquid exiting the hybrid adiabatic cooler and an outlet coupled to the return segment of the outer conduit loop. The direct liquid cooling unit is configured to transfer heat from a second cooling liquid to the first cooling liquid circulating from the inlet of the direct liquid cooling unit to the outlet of the direct liquid cooling unit. In some embodiments, the hybrid adiabatic cooling system includes an inner conduit loop including a first segment fluidly coupled to the feed segment of the outer conduit loop and a second segment fluidly coupled to the return segment of the outer conduit loop. In some embodiments, the hybrid adiabatic cooling system includes an air handling unit disposed inside the data center. The air handling unit includes an inlet fluidly coupled to the first segment of the inner conduit loop and an outlet coupled to the second segment of the inner conduit loop. The air handling unit is configured to transfer heat from indoor ambient air drawn through the air handling unit to the first cooling liquid circulating from the inlet of the air handling unit to the outlet of the air handling unit.

[0015] In some embodiments, the hybrid adiabatic cooling system includes a chiller fluidly coupled to the first and second segments of the inner conduit loop, wherein the chiller is configured to cool the first cooling liquid circulating through the first segment of the inner conduit loop and heat the first cooling liquid circulating through the second segment of the inner conduit loop.

[0016] In some embodiments, the first segment of the inner conduit loop includes a feed line fluidly coupled to the feed segment of the outer conduit loop; a first chiller inlet line fluidly coupled to the feed line and a first inlet of the chiller; a chiller outlet line fluidly coupled to a first outlet of the chiller and the inlet of the air handling unit; and a bypass feed line fluidly coupled to the feed line and the chiller outlet line.

[0017] In some embodiments, the second segment of the inner conduit loop includes a discharge line fluidly coupled to the outlet of the air handling unit; a second chiller inlet line fluidly coupled to the discharge line and a second inlet of the chiller; a second chiller outlet line fluidly coupled to a second outlet of the chiller and the return segment of the outer conduit loop; and a bypass outlet line fluidly coupled to the discharge line and the return segment of the outer conduit loop.

[0018] In some embodiments, the chiller includes an evaporator section disposed between the first inlet and the first outlet of the chiller, wherein the evaporator section is configured to exchange heat between the first cooling liquid and a working fluid such that the working fluid is evaporated to a

gaseous state. In some embodiments, the chiller includes a condenser section disposed between the second inlet and the second outlet of the chiller, wherein the condenser section is configured to exchange heat between the first cooling liquid and the working fluid such that the working fluid is condensed to a liquid state.

[0019] In some embodiments, the hybrid adiabatic cooling system includes a first valve fluidly coupled to the feed line, the first chiller inlet line, and the bypass feed line of the first segment of the inner conduit loop, wherein the first valve is configured to switch between a first position and a second position. When set in the first position, the first valve is configured to direct the first cooling liquid from the feed line to the bypass feed line. When set in the second position the first valve is configured to direct the first cooling liquid from the feed line to the first chiller inlet line.

[0020] In some embodiments, the hybrid adiabatic cooling system includes a controller in communication with the first valve, wherein the controller is configured to set the first valve in the first position when a temperature of the outdoor ambient air is below a first threshold temperature and in the second position when the temperature of the outdoor ambient air is above a first threshold temperature.

[0021] In some embodiments, the hybrid adiabatic cooling system includes a second valve fluidly coupled to the discharge line, the second chiller inlet line, and the bypass outlet line of the second segment of the inner conduit loop, wherein the second valve is configured to switch between a first position and a second position. When set in the first position, the second valve is configured to direct the first cooling liquid from the discharge line to the bypass outlet line. When set in the second position, the second valve is configured to direct the first cooling liquid from the discharge line to the second chiller inlet line.

[0022] In some embodiments, the controller is configured to set the second valve in the first position when the temperature of the outdoor ambient air is below the first threshold temperature and in the second position when the temperature of the outdoor ambient air is above the first threshold temperature.

[0023] In some embodiments, the hybrid adiabatic cooler includes an air inlet; an air outlet; a blower configured to draw the outdoor ambient air from the air inlet to the air outlet; a tube coil disposed between the air inlet and the air outlet, wherein the tube coil is fluidly coupled to the inlet and outlet of the hybrid adiabatic cooler; and an adiabatic pad disposed at the air inlet, wherein the adiabatic pad is configured to moisturize the outdoor ambient air drawn through the air inlet. In some embodiments, the hybrid adiabatic cooler is configured to operate in a dry mode such that the adiabatic pad is dry and in a wet mode such that the adiabatic pad is wet to moisturize the outdoor ambient air drawn through air inlet.

[0024] In some embodiments, the controller is in communication with the hybrid adiabatic cooler, and the controller is configured to set the hybrid adiabatic cooler in the dry mode when the temperature of the outdoor ambient air is below a second threshold temperature and in the wet mode when the temperature of the outdoor ambient air is above the second threshold temperature.

[0025] In some embodiments, the first threshold temperature is in a range from 55° F. to 75° F. (12.8° C. to 23.9° C.), and the second threshold temperature is in a range from 70° F. to 95° F. (21.1° C. to 35.1° C.).

[0026] In some embodiments, the first cooling liquid is water, and the second cooling liquid is a dielectric coolant.

[0027] In some embodiments, the present disclosure provides a hybrid adiabatic cooling system. In some embodiments, the hybrid adiabatic cooling system includes an outer conduit loop including a feed segment and a return segment. In some embodiments, the hybrid adiabatic cooling system includes an inner conduit loop including a first segment fluidly coupled to the feed segment and a second segment fluidly coupled to the return segment of the outer conduit loop. In some embodiments, the hybrid adiabatic cooling system includes a hybrid adiabatic cooler fluidly coupled to the outer conduit loop. The hybrid adiabatic cooler is configured to draw outdoor

ambient air to cool a first cooling liquid circulating from the return segment to the feed segment of the outer conduit loop. In some embodiments, the hybrid adiabatic cooling system includes a direct liquid cooling unit fluidly coupled to the outer conduit loop. The direct liquid cooling unit is configured to transfer heat from a second cooling liquid to the first cooling liquid circulating from the feed segment to the return segment of the outer conduit loop. In some embodiments, the hybrid adiabatic cooling system includes an air handling unit fluidly coupled to the inner conduit loop. The air handling unit is configured to transfer heat from indoor ambient air drawn through the air handling unit to the first cooling liquid circulating from the first segment to the second segment of the inner conduit loop. In some embodiments, the hybrid adiabatic cooling system includes a chiller fluidly coupled to the first segment of the inner conduit loop. The chiller is configured to cool the first cooling liquid circulating through the first segment of the inner conduit loop. In some embodiments, the hybrid adiabatic cooling system includes a controller configured to operate the hybrid adiabatic cooling system in a free cooling mode when a temperature of the outdoor ambient air is less than a first threshold temperature and a chiller mode when the temperature of the outdoor ambient air is greater than a first threshold temperature. In some embodiments, when the hybrid adiabatic cooling system is set in the free cooling mode, the first cooling liquid in the first segment of inner conduit loop bypasses the chiller. In some embodiments, when the hybrid adiabatic cooling system is set in the chiller mode, the first cooling liquid in the first segment of inner conduit loop circulates through the chiller.

[0028] In some embodiments, the hybrid adiabatic cooling system includes a first pump fluidly coupled to the outer conduit loop, the first pump configured to propel the first cooling liquid through the outer conduit loop. In some embodiments, the hybrid adiabatic cooling system includes a second pump fluidly coupled to the first segment of the second conduit loop. The second pump is configured to propel the first cooling liquid circulating in the first segment of the inner conduit loop through the chiller.

[0029] In some embodiments, the controller is in communication with the first pump and the second pump, and the controller is configured to deactivate the second pump when setting the hybrid adiabatic cooling system in the free cooling mode and activate the second pump when setting the hybrid adiabatic cooling system in the chiller mode.

[0030] In some embodiments, the hybrid adiabatic cooler is configured to operate in a dry mode and a wet mode. When set in the dry mode, the hybrid adiabatic does not moisture the outdoor ambient air drawn through the hybrid adiabatic cooler. When set in the wet mode, the hybrid adiabatic cooler moisturizes outdoor ambient air drawn through the hybrid adiabatic cooler.

[0031] In some embodiments, the controller is in communication with the hybrid adiabatic cooler, and the controller is configured to set the hybrid adiabatic cooler in the dry mode when the temperature of the outdoor ambient air is below a second threshold temperature and in the wet mode when the temperature of the outdoor ambient air is above the second threshold temperature. The second threshold temperature is greater than the first threshold temperature.

[0032] In some embodiments, the present disclosure provides a method of controlling a hybrid adiabatic cooling system to cool an interior of a data center. In some embodiments, the method includes a step of cooling, by a hybrid adiabatic cooler, a first cooling liquid circulating from a return segment of an outer conduit loop to a feed segment of the outer conduit loop. In some embodiments, the method includes a step of exchanging, by a direct liquid cooling unit, heat between a second cooling liquid and the first cooling liquid circulating from the feed segment of the outer conduit loop to the return segment of the outer conduit loop. In some embodiments, the method includes a step of diverting a portion of the first cooling liquid circulating in the feed segment of the outer conduit loop to a first segment of an inner conduit loop. In some embodiments, the method includes a step of determining, by a controller, to operate the hybrid adiabatic cooling system in a free cooling mode or a chiller mode based on a temperature of outdoor ambient air drawn through the hybrid adiabatic cooler to cool the first cooling liquid. In

some embodiments, the method includes a step of exchanging, by a room air handling unit, heat between indoor ambient air and the diverted portion of the first cooling liquid circulating from the first segment of the inner conduit loop to a second segment of the inner conduit loop that is fluidly coupled to the return segment of the outer conduit loop. In some embodiments, the diverted portion of the first cooling liquid in the first segment of the inner conduit loop bypasses a chiller when the hybrid adiabatic cooling system is set in the free cooling mode. In some embodiments, the diverted portion of the first cooling liquid in the first segment of the inner conduit loop circulates through the chiller when the hybrid adiabatic cooling system is set in the chiller mode.

[0033] In some embodiments, the controller determines to operate the hybrid adiabatic cooling system in the free cooling mode when the temperature of the outdoor ambient air is less than a first threshold temperature and in the chiller mode when the temperature of the outdoor ambient air is above than a first threshold temperature.

[0034] In some embodiments, the present disclosure provides a controller for a hybrid adiabatic cooling system that includes an outer conduit loop circulating through a data center and an inner conduit loop fluidly coupled to the outer conduit loop. In some embodiments, the controller comprises an input interface configured to receive a temperature signal from an outdoor temperature sensor indicating a temperature of outdoor ambient air. In some embodiments, the controller comprises an output interface configured to transmit an actuation signal to a primary pump to circulate a first cooling liquid from a feed segment of the outer conduit loop to a return segment of the outer conduit loop and through the inner conduit loop fluidly coupled to the feed segment and the return segment of the outer conduit loop. In some embodiments, the outer conduit loop is fluidly coupled to a direct liquid cooling unit disposed in the data center and the inner conduit loop is fluidly coupled to an air handling unit disposed in the data center. In some embodiments, the output interface is configured to transmit an actuation signal to a secondary pump to circulate the first cooling liquid through a chiller fluidly coupled to the inner conduit loop and located upstream of the air handling unit. In some embodiments, the output interface is configured to transmit an actuation signal to a hybrid adiabatic cooler to moisturize the outdoor ambient air drawn through an air inlet of the hybrid adiabatic cooler such that the hybrid adiabatic cooler adiabatically cools the first cooling liquid circulating from the return segment to the feed segment of the outer conduit loop. In some embodiments, the controller comprises a processor in electrical communication with the input interface to receive the temperature signal from the input interface and in electrical communication with the output interface to transmit the actuation signal selectively to the primary pump, the secondary pump, and the hybrid adiabatic cooler. In some embodiments, the processor is configured to operate the hybrid adiabatic cooling system in a: (i) free cooling mode by activating the primary pump and deactivating the secondary pump, (ii) a dry chiller mode by activating the primary pump and the secondary pump, and (iii) a wet chiller mode by activating the primary pump and the secondary pump and actuating the hybrid adiabatic cooler to moisturize the outdoor ambient air. In some embodiments, the processor is configured to determine to operate the hybrid adiabatic cooling system in the free cooling mode, the dry chiller mode, and the wet chiller mode based on the temperature signal received from the outdoor temperature sensor.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0035] The accompanying drawings, which are incorporated herein and form part of the specification, illustrate embodiments and, together with the description, further serve to explain the principles of the embodiments and to enable a person skilled in the relevant art(s) to make and use the embodiments.

[0036] FIG. 1 illustrates a schematic circuit diagram of the hybrid adiabatic cooling system according to an embodiment.

[0037] FIG. 2A illustrates a perspective view of a hybrid adiabatic cooler according to an embodiment.

[0038] FIG. 2B illustrates a cross-sectional view of a hybrid adiabatic cooler according to an embodiment.

[0039] FIG. 3 illustrates a perspective view of a direct liquid cooling unit according to an embodiment.

[0040] FIG. 4 illustrates a perspective view of a direct liquid cooling unit according to an embodiment.

[0041] FIG. 5A illustrates an air handling unit according to an embodiment.

[0042] FIG. 5B illustrates an air handling unit according to an embodiment.

[0043] FIG. 6 illustrates a schematic diagram of a controller for a hybrid adiabatic cooling system according to an embodiment.

[0044] FIG. 7 illustrates a block diagram showing aspects of a method of controlling a hybrid adiabatic cooling system to cool an interior of a data center according to an embodiment.

[0045] The features and advantages of the embodiments will become more apparent from the detail description set forth below when taken in conjunction with the drawings. A person of ordinary skill in the art will recognize that the drawings may use different reference numbers for identical, functionally similar, and/or structurally similar elements, and that different reference numbers do not necessarily indicate distinct embodiments or elements. Likewise, a person of ordinary skill in the art will recognize that functionalities described with respect to one element are equally applicable to functionally similar, and/or structurally similar elements.

DETAILED DESCRIPTION

[0046] Embodiments of the present disclosure are described in detail with reference to embodiments thereof as illustrated in the accompanying drawings. References to “one embodiment,” “an embodiment,” “some embodiments,” “certain embodiments,” etc., indicate that the embodiment described can include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic.

Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it is submitted that it is within the knowledge of one skilled in the art to affect such feature, structure, or characteristic in connection with other embodiments whether or not explicitly described.

[0047] The terms “upstream” and “downstream” as used herein refer to the relative location of a component (e.g., cooling unit) with respect to the direction of fluid flow. For example, the term “upstream,” as used herein, refers to a relative position of a component in a conduit loop that is not as far along in the direction of the fluid flow. The term “downstream,” as used herein, refers to a relative position in a conduit loop that is farther along in the direction of fluid flow.

[0048] The following examples are illustrative, but not limiting, of the present embodiments. Other suitable modifications and adaptations of the variety of conditions and parameters normally encountered in the field, and which would be apparent to those skilled in the art, are within the spirit and scope of the disclosure.

[0049] Thermal management of electrical equipment in data centers is essential for maintaining operability. Conventional data center's cooling systems routinely condition the air of the data center's interior space to a predetermined temperature and directly cool the data center's electric components through thermal conduction or immersive cooling with a coolant. However, a significant amount of energy is consumed to maintain operable conditions (temperature and humidity) for data centers, particular facilities including a high concentration of computing equipment. Thus, there is a need for a data center cooling system that can cool the interior air of the data center and direct liquid cooling of the electronic components, while consuming lower energy

and less water.

[0050] According to embodiments described herein, a hybrid adiabatic cooling system of the present disclosure can overcome one or more of these deficiencies, for example, by providing a closed loop of a first cooling liquid (e.g. water) that directs the first cooling liquid to two distinct heat loads of the data center at two different temperatures. The hybrid adiabatic cooling system includes an inner conduit loop and an outer conduit loop fluidly coupled to the inner conduit loop. The inner conduit loop supplies the first cooling liquid to a computer room air handling unit that generates cool air to convectively cool electric equipment in the data center. The outer conduit loop supplies the first cooling liquid to direct liquid cooling unit that uses a second cooling liquid to directly cool the electric equipment of the data center. The hybrid adiabatic cooling system includes a hybrid adiabatic cooler fluidly coupled to the outer conduit loop to cool the first cooling liquid returned from the heat loads in the data center. When the outdoor ambient temperature is below a first threshold temperature (e.g., 65.0° F., 18.3° C.), the hybrid adiabatic cooler supplies chilled first cooling liquid to the outer and inner conduit loops at an operating temperature without the use of mechanical refrigeration to cool the first cooling liquid, thereby reducing the energy consumption of the cooling system.

[0051] FIG. 1 illustrates a hybrid adiabatic cooling system **100** according to an embodiment. Hybrid adiabatic cooling system **100** circulates a first cooling liquid, such as water, in a closed loop through a data center **10**—a building having a confined space for housing a network of computer systems, servers, telecommunication devices, and servers—to assist the thermal management of electronic components stored in data center **10**. Hybrid adiabatic cooling system **100** may supply the first cooling liquid at two different temperatures to two different types of cooling units—for example, a direct liquid cooling unit and an air handling unit—located in data center **10** such that the hybrid adiabatic cooling system **100** dissipates heat from the data center's electronic components by two modes of heat exchange: direct liquid cooling the electronic components and air conditioning the internal space of data center **10**.

[0052] In some embodiments, hybrid adiabatic cooling system **100** may include an outer conduit loop **110** to direct chilled first cooling liquid (e.g., water) to one or more cooling units (e.g., a direct liquid cooling unit) located in data center **10** and return heated first cooling liquid discharged from the one or more cooling units out of data center **10**. For example, outer conduit loop **110** may include a feed segment **112** to direct the chilled first cooling liquid from the outside of data center **10** to a first set of cooling units (e.g., a direct liquid cooling unit **140**) located inside of data center **10**. Outer conduit loop **110** may include a return segment **114** to direct the heated first cooling liquid discharged from the one or more cooling units to outside of data center **10** so that heat in the first cooling liquid may be dissipated to the outdoor ambient air. Feed segment **112** and return segment **114** of outer conduit loop **110** may include a network of pipes, tubes, conduits, hoses, and/or fluid lines suitable for circulating the first cooling liquid through outer conduit loop **110**. In some embodiments, feed segment **112** and return segment **114** may each consist of a single pipe, tube, or conduit for circulating the first cooling liquid.

[0053] In some embodiments, hybrid adiabatic cooling system **100** may include an inner conduit loop **120** to direct the chilled first cooling liquid from feed segment **112** of outer conduit loop **110** to one or more cooling units (e.g., an air handling unit) located in data center **10** and direct the heated first cooling liquid discharged from the one or more cooling units to return segment **114** of outer conduit loop **110**. For example, inner conduit loop **120** may include a first segment **122** fluidly coupled to feed segment **112** of outer conduit loop **110** and a second set of cooling units (e.g., air handling unit **150**) that are distinct from the first set of cooling units. Inner conduit loop **120** may include a second segment **124** fluidly coupled to the second set of cooling units and return segment **114** of outer conduit loop **110**. First segment **122** and second segment **124** of inner conduit loop **120** may include a network of pipes, tubes, conduits, hoses, and/or fluid lines suitable for circulating the first cooling liquid through inner conduit loop **120**.

[0054] In some embodiments, hybrid adiabatic cooling system **100** may include a hybrid adiabatic cooler **130** located outside of data center **10**. Hybrid adiabatic cooler **130** may be fluidly coupled to outer conduit loop **110**. Hybrid adiabatic cooler **130** may cool the first cooling liquid (e.g., water) circulating through outer conduit loop **110** by dissipating heat from the first cooling liquid to the outdoor ambient air. Hybrid adiabatic cooler **130** may include an inlet **132** fluidly coupled to return segment **114** of outer conduit loop **110** to receive the heated first cooling liquid discharged from the one or more cooling units (e.g., a direct liquid cooling unit **140** and an air handling unit **150**) located in data center **10**. Hybrid adiabatic cooler **130** may include an outlet **134** fluidly coupled to feed segment **112** of outer conduit loop **110** to supply cooled first cooling liquid to the one or more cooling units located in data center **10**.

[0055] Hybrid adiabatic cooler **130** may operate in two modes: a dry mode and a wet mode. When hybrid adiabatic cooler **130** is set in the dry mode, the outdoor ambient air is drawn through hybrid adiabatic cooler **130** without moisturizing the air intake such that the dry-bulb temperature of the air is not cooled before dissipating heat from the first cooling liquid. When hybrid adiabatic cooler **130** is set in a wet mode, the outdoor ambient air drawn through hybrid adiabatic cooler **130** is moisturized such that the air intake is cooled under the concept of adiabatic evaporative cooling before dissipating heat from the first cooling liquid. During wet mode, the moisture added to the outdoor ambient air drawn through the hybrid adiabatic cooler **130** is evaporated to cool the dry-bulb temperature of the air to its wet-bulb temperature as the enthalpy of moisten air remains constant. The adiabatically cooled air drawn through hybrid adiabatic cooler **130** during wet mode will have a higher humidity level and a lower dry-bulb temperature than the non-moisturized air drawn through hybrid adiabatic cooler **130** during dry mode, thereby dissipating heat from the first cooling liquid at a higher rate.

[0056] FIGS. 2A and 2B illustrates one embodiment of hybrid adiabatic cooler **130**. Hybrid adiabatic cooler **130** may include a housing **200** defining an air inlet **210** for receiving outdoor ambient air and an air outlet **220** for discharging outdoor ambient air. Hybrid adiabatic cooler **130** may include a blower to draw the outdoor ambient air from air inlet **210** to air outlet **220**. In some embodiments, the fan speed of blower may be adjustable to control the flow rate of air intake. Hybrid adiabatic cooler **130** may include a heat exchanger **240** disposed between air inlet **210** and air outlet **220**. Heat exchanger **240** may be fluidly coupled to inlet **132** and outlet **134** of hybrid adiabatic cooler **130** to circulate the first cooling liquid (e.g., water) from return segment **114** of outer conduit loop **110** to feed segment **112** of outer conduit loop **110**. In some embodiments, heat exchanger **240** may include an inlet header **242**, an outlet header **244**, and a plurality of tube bundles **246** fluidly coupled to inlet header **242** and outlet header **244**. Hybrid adiabatic cooler **130** may include an adiabatic pad **250** disposed proximate to air inlet **210** to moisturize air drawn through air inlet **210**. In some embodiments, adiabatic pad **250** may include an open-mesh panel, such as a matrix of glass fiber and ceramic composite materials, which is configured to be saturated with water to moisturize air passing therethrough.

[0057] With reference to FIG. 1, in some embodiments, hybrid adiabatic cooling system **100** may include a direct liquid cooling unit **140** located inside data center **10** and fluidly coupled to outer conduit loop **110**. Direct liquid cooling unit **140** exchanges heat between a second cooling liquid (e.g., a coolant) directed to cooling electric components of data center **10** and the first cooling liquid (e.g., water) circulating through outer conduit loop **110**. By cooling the second cooling liquid, direct liquid cooling unit **140** heats the first cooling liquid circulating from feed segment **112** to return segment **114** of outer conduit loop **110**. Direct liquid cooling unit **140** may include an inlet **142** fluidly coupled to feed segment **112** of outer conduit loop **110** to receive the chilled first cooling liquid (e.g., water) supplied from outlet **134** of hybrid adiabatic cooler **130**. Direct liquid cooling unit **140** may include an outlet **144** fluidly coupled to return segment **114** of outer conduit loop **110** to direct the heated first cooling liquid to inlet **132** of hybrid adiabatic cooler **130**.

[0058] FIG. 3 illustrates a direct liquid cooling unit **300** according to one embodiment. Direct

liquid cooling unit **300** may cool electric components **20** of data center **10** by directly contacting electric components **20** with a dielectric coolant used as the second cooling liquid. For example, as shown in FIG. **3**, direct liquid cooling unit **300** may include a housing **310** defining a tank **320** filled with dielectric coolant to immerse electric components **20**. Direct liquid cooling unit **300** may include a heat exchanger disposed in housing **310** to exchange heat between the second cooling liquid (e.g., dielectric coolant) and the first cooling liquid (e.g., water) circulating through direct liquid cooling unit **300**. In some embodiments, the second cooling liquid may include perfluorinated carbons and/or polyalphaolefin hydrocarbons.

[0059] FIG. **4** illustrates a direct liquid cooling unit **400** according to one embodiment. Direct liquid cooling unit **400** may cool electric components **20** of data center **10** by indirectly contacting the electric components **20** with the second cooling liquid through a cold plate **430**. For example, as shown in FIG. **4**, direct liquid cooling unit **400** may include a housing **410** having a plurality of racks **412** holding electric components **20** and a coolant loop **420** fluidly coupled to cold plates **430**, which are in thermal contact with electric components **20**. Direct liquid cooling unit **400** may circulate a second cooling liquid **422** (e.g., coolant) through coolant loop **420** to direct second cooling liquid **422** through cold plates **430**. Direct liquid cooling unit **400** may include a heat exchanger **440** disposed in housing **410** to exchange heat between second cooling fluid **422** and a first cooling liquid **450** (e.g., water) circulating through direct liquid cooling unit **400**.

[0060] With reference to FIG. **1**, in some embodiments, hybrid adiabatic cooling system **100** may include an air handling unit **150** located inside data center **10** and fluidly coupled to inner conduit loop **120**. Air handling unit **150** exchanges heat between indoor ambient air of data center **10** and the first cooling liquid (e.g., water) circulating through inner conduit loop **120**. In cooling the indoor ambient air of data center **10**, air handling unit **150** heats the first cooling liquid while circulating from first segment **122** to second segment **124** of inner conduit loop **120**. Air handling unit **150** may include an inlet **152** fluidly coupled to first segment **122** of inner conduit loop **120** to receive the chilled first cooling liquid diverted from feed segment **112** of inner conduit loop **110**. Air handling unit **150** may include an outlet **154** fluidly coupled to second segment **124** of inner conduit loop **120** to direct the diverted portion of heated first cooling liquid to return segment **114** of outer conduit loop **110**.

[0061] FIG. **5A** illustrates one embodiment of an air handling unit **500**. Air handling unit **500** may include a housing **510**. Housing **510** of air handling unit **500** may define an air inlet **512** for receiving indoor ambient air of a room **12** in data center **10** and an air outlet **520** for discharging conditioned air. In some embodiments, as shown in FIG. **5A**, air outlet **520** opens into a plenum **30** defined between a room floor **40** and a base floor **50** of data center **10** that defines an air passage opening into aisles defined between rows of racks **60**, where each rack **60** houses electric components **20** of data center **10**. Air handling unit **500** may include a fan to draw heated air exhausted from racks **60** through air inlet **512**. Air handling unit **500** may include a heat exchanger **530** disposed between air inlet **512** and air outlet **520**. Heat exchanger **530** may be fluidly coupled to an inlet **502** and an outlet **504** of air handling unit **500** to circulate the first cooling liquid from first segment **122** of inner conduit loop **120** to second segment **124** of inner conduit loop **120**. Heat exchanger **530** cools the air drawn through air handling unit **500** such that first cooling liquid (e.g., water) is heated.

[0062] FIG. **5B** illustrates another embodiment of an air handling unit **550**. Air handling unit **550** may include a housing **560**. Housing **560** of air handling unit **550** may define an air inlet **562** for receiving indoor ambient air of room **12** in data center **10** and an air outlet **564** for discharging cooled air. In some embodiments, as shown in FIG. **5B**, air outlet **564** opens directly into room **12** of data center **10** to direct cooled air between rows of racks **60** of data center **10**. Air handling unit **550** may include one or more fans **580** to draw heated air exhausted from racks **60** through air inlet **562**. Air handling unit **550** may include a filter **566** disposed at air inlet **562** to remove particulates from the heated air recirculated through air handling unit **550**. Air handling unit **550** may include a

heat exchanger **570** disposed between air inlet **562** and air outlet **564**. Heat exchanger **570** may be fluidly coupled to an inlet **552** and an outlet **554** of air handling unit **550** to circulate the first cooling liquid from first segment **122** of inner conduit loop **120** to second segment **124** of inner conduit loop **120**. Heat exchanger **570** cools the air drawn through air handling unit **550** such that first cooling liquid (e.g., water) is heated.

[0063] In some embodiments, hybrid adiabatic cooling system **100** may include a chiller **160** fluidly coupled to inner conduit loop **120**. Chiller **160** may circulate a working fluid (e.g., a refrigerant in a refrigerant cycle or water) to cool the first cooling liquid circulating through first segment **122** of inner conduit loop **120** and to heat the first cooling liquid circulating through second segment **124** of the inner conduit loop **120**. For example, chiller **160** may include an evaporator section **163** having a bundle of tubes to transfer heat from the first cooling liquid circulating through first segment **122** of inner conduit loop **120** to the liquid working fluid such that the liquid working fluid is evaporated to a gaseous state. Evaporator section **163** of chiller **160** may include a first inlet **161** fluidly coupled to first segment **122** of inner conduit loop **120** and a first outlet **162** fluidly coupled to first segment **122** of inner conduit loop **120** to allow the first cooling liquid to circulate through evaporator section **163** of chiller **160**. Chiller **160** may include a condenser section **166** having a bundle of tubes to transfer heat from the gaseous working fluid to the first cooling liquid circulating through second segment **124** of inner conduit loop **120** such that the gaseous working fluid is condensed to a liquid state. Condenser section **166** of chiller **160** may include a second inlet **164** fluidly coupled to second segment **124** of inner conduit loop **120** and a second outlet **165** fluidly coupled to second segment **124** of inner conduit loop **120** to circulate the first cooling liquid through condenser section **166** of chiller **160**.

[0064] In some embodiments, first segment **122** and second segment **124** of inner conduit loop **120** may include a combination of fluid lines (e.g., pipes, tube, conduits) and one or more valves to selectively allow the first cooling liquid (e.g., water) to bypass chiller **160**. In bypassing chiller **160**, the first cooling liquid circulates through all the data center's cooling units (e.g., direct liquid cooling unit **140**, air handling unit **150**) without the use of mechanical refrigeration ("free cooling").

[0065] For example, in some embodiments, first segment **122** of inner conduit loop **120** may include a feed line **171** fluidly coupled to feed segment **112** of outer conduit loop **110** to divert the chilled first cooling liquid from feed segment **112** toward air handling unit **150**. First segment **122** of inner conduit loop **120** may include a first chiller inlet line **172** fluidly coupled to feed line **171** and first inlet **161** of chiller **160** to direct the diverted portion of the first cooling liquid to evaporator section **163** of chiller **160**. First segment **122** of inner conduit loop **120** may include a first chiller outlet line **173** fluidly coupled to first outlet **162** of chiller **160** and inlet **152** of air handling unit **150** to direct the diverted portion of the first cooling liquid from evaporator section **163** of chiller **160** to air handling unit **150**. First segment **122** of inner conduit loop **120** may include a bypass feed line **174** fluidly coupled to feed line **171** and first chiller outlet line **173** such that diverted portion of the first cooling liquid bypasses evaporator section **163** of chiller **160** while circulating to inlet **152** of air handling unit **150**.

[0066] In some embodiments, inner conduit loop **120** may include a first valve **126** fluidly coupled to feed line **171**, first chiller inlet line **172**, and bypass feed line **174** of first segment **122** of the inner conduit loop **120**. First valve **126** switches between a first position and a second position to alternate the direction of the first cooling liquid circulating through first segment **122** of inner conduit loop **120**. For example, when set in the first position, first valve **126** directs the first cooling liquid from feed line **171** to bypass feed line **174** such that the portion of the first cooling liquid diverted from feed segment **112** of outer conduit loop **110** to air handling unit **150** bypasses evaporator section **163** of chiller **160**. When set in the second position, first valve **126** directs the first cooling liquid from feed line **171** to first chiller inlet line **172** to circulate the first cooling liquid circulates through evaporator section **163** of chiller **160**, such that the first cooling liquid is

cooled by evaporator section **163** of chiller **160** (e.g., by mechanical refrigeration) before reaching first inlet **152** of air handling unit **150**. In some embodiments, first segment **122** of inner conduit loop **120** may include two two-way valves, rather than one three-way valve, to selectively direct the first cooling liquid from feed line **171** to either first chiller inlet line **172** or bypass feed line **174**.

[0067] In some embodiments, second segment **124** of inner conduit loop **120** may include a discharge line **175** fluidly coupled to outlet **154** of air handling unit **150** to receive the diverted portion of the first cooling liquid discharged from air handling unit **150**. Second segment **124** of inner conduit loop **120** may include a second chiller inlet line **176** fluidly coupled to discharge line **175** and second inlet **164** of chiller **160** to direct the diverted portion of the first cooling liquid to condenser section **166** of chiller **160**. Second segment **124** of inner conduit loop **120** may include a second chiller outlet line **177** fluidly coupled to second outlet **165** of chiller **160** and return segment **114** of outer conduit loop **110** such that diverted portion of the first cooling liquid discharged from condenser section **166** of chiller is merged back to the first cooling liquid circulating through return segment **114** of outer conduit loop **110**. Second segment **124** of inner conduit loop **120** may include a bypass outlet line **178** fluidly coupled to discharge line **175** and return segment **114** of outer conduit loop **110** such that the diverted portion of first cooling liquid discharged from air handling unit **150** bypasses condenser section **166** of chiller **160** while merging back with the first cooling liquid circulating through return segment **114** of outer conduit loop **110**.

[0068] In some embodiments, inner conduit loop **120** may include a second valve **128** fluidly coupled to discharge line **175**, second chiller inlet line **176**, and bypass outlet line **178** of second segment **124** of inner conduit loop **120**. Second valve **128** switches between a first position and a second position to alternate the direction of the first cooling liquid circulating through second segment **124** of inner conduit loop **120**. For example, when set in the first position, second valve **128** directs the first cooling liquid from discharge line **175** to bypass outlet line **178** such that first cooling liquid bypasses condenser section **166** of chiller **160**. When set in the second position, second valve **128** directs the first cooling liquid from discharge line **175** to second chiller inlet line **176** such that the first cooling liquid is further heated by condenser section **166** of chiller **160** (e.g., by mechanical refrigeration) before merging with the first cooling liquid circulating through return segment **114** of outer conduit loop **110**. In some embodiments, second segment **124** of inner conduit loop **120** may include two two-way valves, rather than one three-way valve, to selectively direct the first cooling liquid from discharge line **175** to either second chiller inlet line **176** or bypass outline line **178**.

[0069] In some embodiments, hybrid adiabatic cooling system **100** may include a primary pump **190** fluidly coupled to outer conduit loop **110** to propel the first cooling liquid through outer conduit loop **110**. Hybrid adiabatic cooling system **100** may include a secondary pump **192** fluidly coupled to inner conduit loop **120** to propel the first cooling liquid through inner conduit loop **120**, including through chiller **160**.

[0070] While FIG. **1** illustrates one hybrid adiabatic cooler **130**, one direct liquid cooling unit **140**, one air handling unit **150**, one chiller **160**, one primary pump **190**, and one secondary pump **192**, hybrid adiabatic cooling system **100** may include a plurality of hybrid adiabatic coolers, direct liquid cooling units, air handling units, chillers, primary pumps, and secondary pumps without departing from the scope of the present disclosure.

[0071] While FIG. **1** illustrates direct liquid cooling unit **140** fluidly coupled to outer conduit loop **110** and air handling unit **150** and chiller **160** fluidly coupled to inner conduit loop **120**, in other embodiments, hybrid adiabatic cooling system **100** may include a direct liquid cooling unit fluidly coupled to the inner conduit loop and an air handling unit and a chiller fluidly coupled to the outer conduit loop without departing from the scope of the present disclosure.

[0072] With reference to FIG. **6**, hybrid adiabatic cooling system **100** may include a controller **600** in electrical communication, such as through power line communication or BACnet

communication PLC or other analog or digital control system, with first valve **126**, second valve **128**, hybrid adiabatic cooler **130**, direct liquid cooling unit **140**, air handling unit **150**, chiller **160**, primary pump **190**, and/or secondary pump **192**. For example, controller **600** may include an output interface configured to transmit control signals to first valve **126**, second valve **128**, hybrid adiabatic cooler **130**, primary pump **190**, and secondary pump **192** such that controller **600** is configured to actuate first valve **126**, second valve **128**, hybrid adiabatic cooler **130**, primary pump **190**, and secondary pump **192**. In some embodiments, controller **600** may be incorporated as a module of a controller in the data center's building automation system.

[0073] In some embodiments, controller **600** may include an input interface to receive input signals from one or more sensors and/or data sources to monitor the conditions (e.g., dry-bulb temperature) of outdoor ambient air, indoor air of data center **10**, and/or the first cooling liquid. For example, hybrid adiabatic cooling system **100** may include an outdoor temperature sensor **610** disposed outside of data center **10**, such as being proximate to hybrid adiabatic cooler **130**. Outdoor temperature sensor **610** (e.g., a thermocouple, thermometer, thermistor) may detect the temperature of outdoor ambient air being drawn into hybrid adiabatic cooler **130** and generate an input signal to controller **600** indicating the temperature of outdoor ambient air being drawn into hybrid adiabatic cooler **130**.

[0074] Hybrid adiabatic cooling system **100** may include an indoor temperature sensor **620** disposed in the data center **10**, for example, being proximate to air handling unit **150**. Indoor temperature sensor **620** (e.g., a thermocouple, thermometer, thermistor) may detect the temperature of indoor ambient air being drawn into air handling unit **150** and generate an input signal to controller **600** indicating the temperature of indoor ambient air being drawn into air handling unit **150**.

[0075] Hybrid adiabatic cooling system **100** may include one or more cooling liquid sensors **630** disposed in outer conduit loop **110**, such as in feed segment **112**, and/or in inner conduit loop **120**, such as in first segment **122**. Cooling liquid sensor **630** (e.g., a thermocouple, thermometer, thermistor) may detect the temperature of the first cooling liquid circulating through feed segment **112** of outer conduit loop **110** and/or the temperature of the first cooling liquid circulating through first segment **122** of inner conduit loop **120**. Cooling liquid sensor **630** may generate an input signal to controller **600** indicating the temperature of the first cooling liquid circulating through feed segment **112** of outer conduit loop **110** and/or the temperature of the first cooling liquid circulating first segment **122** of inner conduit loop **120**.

[0076] Hybrid adiabatic cooling system **100** may include one or more additional data sources **640**, such as receiving input from a network (e.g., local area network, wide area network). For example, data source **640** may indicate weather forecasts that can be processed to determine a predicted outdoor ambient temperature and humidity.

[0077] In some embodiments, controller **600** may include a processor (e.g., a microprocessor, a multi-core processor, a central processing unit) configured to receive input signals from outdoor temperature sensor **610**, indoor temperature sensor **620**, cooling liquid sensor **630**, and data source **640** and generate actuation signals transmitted to first valve **126**, second valve **128**, hybrid adiabatic cooler **130**, primary pump **190**, and secondary pump **192** to control the flow and temperature of the first cooling liquid through outer conduit loop **110** and inner conduit loop **120**. Controller **600** may include memory comprising computer storage media in the form of volatile memory, such as RAM, and/or nonvolatile memory, such as ROM. In some embodiments, the memory of controller **600** may store computer readable instructions, data structures, program modules, and other data, which are inputted to the processor for the execution of operations, as described herein. Controller **600** can include any type of circuitry components, such as a bus, for transmitting instructions stored in the memory to the processor.

[0078] In operation, controller **600** is configured to selectively actuate first valve **126**, second valve **128**, hybrid adiabatic cooler **130**, direct liquid cooling unit **140**, air handling unit **150**, chiller **160**,

primary pump **190**, and/or secondary pump **192** based on input from outdoor temperature sensor **610**, indoor temperature sensor **620**, cooling liquid sensor **630**, and data source **640** according to a predetermined protocol that includes a plurality of operation modes. The plurality of operation modes can include, for example, a free cooling mode, a dry chiller mode, and a wet chiller mode. Free Cooling Mode (e.g., Low Ambient Outdoor Less Than 65° F., 18.3° C.)

[0079] When hybrid adiabatic cooling system **100** is operating in the free cooling mode, first valve **126** and second valve **128** are each set in the first position and secondary pump **192** remains in an off state, such that first cooling liquid diverted from feed segment **112** of outer conduit loop **110** to inner conduit loop **120** bypasses chiller **160**. Hybrid adiabatic cooler **130** operates in dry mode during free cooling mode, thereby not moisturizing outdoor ambient air drawn through hybrid adiabatic cooler **130**.

[0080] Hybrid adiabatic cooling system **100** is configured to operate in the free cooling mode when the outdoor ambient air temperature is lower than a first threshold temperature, for example in a range from 55° F. to 75° F. (12.8° C. to 23.9° C.), such as 65° F. (18.3° C.). When the outdoor ambient temperature is below the first temperature threshold, hybrid adiabatic cooler **130** is configured to cool the heated first cooling liquid circulated received return segment **114** to a desired operating temperature (e.g., 22.8° C., 73.0° F.) sufficient for allowing direct liquid cooling unit **140** and room air handling unit **150** to dissipate their target heat loads in the data center **10** without using chiller **160** or any refrigeration to cool the first cooling liquid (e.g., water). The first threshold temperature and/or desired operating temperature may vary based on the heat load of the data center **10** and heat dissipation capacity of the hybrid adiabatic cooler **130**, direct liquid cooling unit **140**, and air handling unit **150**.

[0081] When receiving the first cooling liquid at the desired operating temperature, air handling unit **150** emits conditioned indoor air at a predetermined temperature (e.g., 26.7° C., 80.0° F.) to convectively air cool electric components **20** of data center **10**. In doing so, air handling unit **150** maintains a lower temperature differential Δ (e.g., 3.9° C., 7.0° F.) between the temperature of first cooling liquid (e.g., water) entering inlet **152** and the temperature of the emitted condition air compared to temperature differentials Δ (e.g., 5.5° C., 10.0° F.) maintained in traditional, prior art computer room air handling units. By maintaining this decreased temperature differential Δ between the first cooling liquid and conditioned air, the hybrid adiabatic cooler **130** increases the hours per year of the cooling system **100** operating in free cooling mode (cooling without the use of mechanical refrigeration), thereby decreasing the pump energy demand by approximately 50% and other costs, including the reduction or elimination of pipe insulation.

Dry Chiller Mode (e.g., Ambient Outdoor Equal to or Greater Than 65° F., 18.3° C.)

[0082] When hybrid adiabatic cooling system **100** is operating in the dry chiller mode, first valve **126** and second valve **128** are each set in the second position and secondary pump **192** is activated to an operational state, such that first cooling liquid diverted from feed segment **112** of outer conduit loop **110** to inner conduit loop **120** is circulated through chiller **160** to further cool the diverted portion of first cooling liquid before supplied to air handling unit **150**. Hybrid adiabatic cooler **130** operates in dry mode during dry chiller mode, thereby not moisturizing outdoor ambient air drawn through hybrid adiabatic cooler **130**.

[0083] Hybrid adiabatic cooling system **100** is configured to operate in the dry chiller mode when the outdoor ambient temperature is equal to above the first threshold temperature (e.g., 18.3° C., 65.0° F.) and below a second threshold temperature for example in a range from 70° F. to 95° F. (12.8° C. to 23.9° C.), such as 82° F. (27.9° C.). When the outdoor ambient temperature is above the first temperature threshold, hybrid adiabatic cooler **130** is not configured to cool the heated first cooling liquid received from return segment **114** of outer conduit loop **110** to the desired operating temperature (e.g., 22.8° C., 73.0° F.), and the first cooling liquid supply temperature floats up in response to a rise in outdoor ambient conditions.

[0084] As the temperature of the first cooling liquid (e.g., water) rises above the desired operating

temperature (e.g., 22.8° C., 73.0° F.), controller **600** of hybrid adiabatic cooling system **100** responds to maintain the first cooling liquid temperature delivered to inner conduit loop **120** by actuating secondary pump **192** in series with the working fluid cooled evaporator section **163** of chiller **160** cooled by a working fluid (e.g., water or refrigerant), thereby lowering first cooling liquid temperature in inner conduit loop **120** to the desired operating temperature (e.g., 22.8° C., 73.0° F.). In some embodiments, during dry chiller mode, the first cooling liquid is discharged from air handling unit **150** at a nominal temperature (e.g., 33.9° C., 93.0° F.), then passes through condenser section **166** of chiller **160**, thereby heating the first cooling liquid to a nominal temperature (e.g., 44.4° C., 112.0° F.) before merging with the first cooling liquid circulating through return segment **114** of outer conduit loop **110**.

Wet Chiller Mode (e.g., Ambient Outdoor Equal to or Greater Than 82° F., 27.9° C.)

[0085] When hybrid adiabatic cooling system **100** is operating in the wet chiller mode, first valve **126** and second valve **128** are each set in the second position and secondary pump **192** is activated to an operational state, such that first cooling liquid diverted from feed segment **112** of outer conduit loop **110** to inner conduit loop **120** is circulated through chiller **160** to further cool the diverted portion of first cooling liquid before supplied to air handling unit **150**. Furthermore, hybrid adiabatic cooler **130** operates in the wet mode during wet chiller mode, thereby moisturizing outdoor ambient air drawn through hybrid adiabatic cooler **130**.

[0086] Hybrid adiabatic cooling system **100** is configured to operate in the wet chiller mode when the outdoor ambient temperature is equal to above the second threshold temperature (e.g., 27.8° C., 82.0° F.). When the outdoor ambient temperature is equal to the second temperature threshold, hybrid adiabatic cooler **130** is configured to supply the first cooling liquid through feed segment **112** of outer conduit loop **110** at a maximum operating temperature (e.g., 32.2° C., 90.0° F.) adequate for allowing the direct liquid cooling unit **140** to dissipate its target heat load in data center **10**. The maximum operating temperature may vary based on the heat load of data center **10** and the heat dissipating capacity of adiabatic cooler **130** and/or direct liquid cooling unit **140**.

[0087] However, when the outdoor ambient temperature is above the second temperature threshold, hybrid adiabatic cooler **130** is not configured to supply the first cooling liquid through feed segment **112** of outer conduit loop **110** at a maximum operating temperature (e.g., 32.2° C., 90.0° F.) such that the first cooling liquid temperature potentially rises above the maximum operating temperature. In response, controller **600** of hybrid adiabatic cooling system **100** actuates hybrid adiabatic cooler **130** to operate in the wet mode (e.g., adiabatic pad **250** moisturizes air intake) to pre-cool the outdoor ambient air drawn through hybrid adiabatic cooler **130**, thereby increasing the heat dissipation rate of the first cooling liquid circulating through hybrid adiabatic cooler **130**. By operating in the wet mode, hybrid adiabatic cooler **130** maintains the first cooling liquid circulating through feed segment **112** at the maximum operating temperature (e.g., 32.2° C., 90.0° F.). In some embodiments, the first cooling liquid discharged from direct liquid cooling unit **140** into return segment **114** is combined with the diverted portion of the first cooling liquid circulating through inner conduit loop **120**, each set at a nominal temperature (e.g., 44.4° C., 112.0° F.).

Method of Operation

[0088] FIG. 7 illustrates an example method **700** of controlling hybrid adiabatic cooling system **100** to cool the interior of data center **10**.

[0089] In some embodiments, method **700** may include a step **710** of cooling the first cooling liquid (e.g., water) circulating from return segment **114** of outer conduit loop **110** to feed segment **112** of outer conduit loop **110**. Step **710** may include using hybrid adiabatic cooler **130** to cool the first cooling liquid. For example, step **710** may include activating the hybrid adiabatic cooler's blower to draw ambient air through air inlet **210** and circulating the first cooling liquid (e.g., water) through heat exchanger **240**, thereby cooling the first cooling liquid. Step **710** may include actuating primary pump **190** to propel the first cooling liquid through outer conduit loop **110**. Step

710 may include setting first valve **126** and second valve **128** in the first position and setting chiller **160** in an off state. Step **710** may result in cooling the first cooling liquid in a temperature range from a desired operating temperature (e.g., 22.8° C., 73.0° F.) to a maximum operating temperature (e.g., 32.2° C., 90.0° F.).

[0090] In some embodiments, method **700** may include a step **720** of exchanging heat between the second cooling liquid (e.g., coolant) and the first cooling liquid (e.g., water) circulating from feed segment **112** of the outer conduit loop **110** to return segment **114** of outer conduit loop **110**. Step **720** may include using direct liquid cooling unit **140** to exchange heat between the first cooling liquid and the second cooling liquid. Step **720** may include using cold plates cooled by the second cooling liquid of the direct liquid cooling unit **140** to dissipate heat from electric components **20** of data center **10**. Step **720** may include immersing electric components **20** with a dielectric coolant, used as the second cooling liquid in direct liquid cooling unit **140**, to dissipate heat from electric components **20** of data center **10**. Step **720** may result in heating the first cooling liquid to a nominal temperature (e.g., 44.4° C., 112.0° F.).

[0091] In some embodiments, method **700** may include a step **730** of diverting a portion of the first cooling liquid circulating in feed segment **112** of the outer conduit loop **110** to first segment **122** of inner conduit loop **120**. Step **730** may include diverting the first cooling liquid to feed line **171** of first segment **122** of inner conduit loop **120**.

[0092] In some embodiments, method **700** may include a step **740** of determining to operate the hybrid adiabatic cooling system **100** in a free cooling mode or a chiller mode (e.g., dry chiller mode or wet chiller mode) based on a temperature of outdoor ambient air drawn through hybrid adiabatic cooler **130** to cool the first cooling liquid. Step **740** may include using controller **600** to determine the mode of operation for hybrid adiabatic cooling system **100**. For example, step **740** may include using controller **600** to determine whether the outdoor ambient air temperature is above the first threshold temperature (65° F.). Step **740** may include receiving an input signal from outdoor temperature sensor **610** to determine the outdoor ambient air temperature. Step **740** may include using an algorithm stored in the memory of controller **600** to determine the mode of operation based on the current outdoor ambient air temperature, other weather conditions (e.g., humidity, forecasted temperature), temperature of the first cooling liquid, and/or heat load of data center as inputs.

[0093] In some embodiments, method **700** may include a step **750** of operating hybrid adiabatic cooling system **100** in a free cooling mode such that the diverted portion first cooling liquid circulating through inner conduit loop **120** bypasses chiller **160**. Step **750** may include setting or maintaining first valve **126** and second valve **128** in the first position. Step **750** may include setting or maintaining secondary pump **192** and chiller **160** in an off state. Step **750** may include using controller **600** to control positioning of first valve **126** and second valve **128** and operation state of chiller **160** and secondary pump **192**. Step **750** may result in the first cooling liquid circulating from feed line **171** to bypass feed line **174** of first segment **122** of inner conduit loop **120**.

[0094] In some embodiments, while operating hybrid adiabatic cooling system **100** in the free cooling mode, method **700** may include a step **755** of exchanging heat between indoor ambient air and the diverted portion of the first cooling liquid circulating from first segment **122** to second segment **124** of inner conduit loop **120**. Step **765** may include using the air handling unit **150** to exchange between indoor ambient air and the diverted portion of the first cooling liquid.

[0095] In some embodiments, method **700** may include a step **760** of operating hybrid adiabatic cooling system **100** in a chiller mode such that diverted portion of first cooling liquid circulates through chiller **160**. Step **760** may include setting first valve **126** and second valve **128** in the second position. Step **760** may include actuating secondary pump **192** and chiller **160**. Step **760** may include using controller **600** to control positioning of first valve **126** and second valve **128** and operation state of chiller **160** and secondary pump **192**. Step **750** may result in the first cooling liquid circulating from feed line **171** to first chiller inlet line **172** and then through evaporator

section **163** of chiller **160** before being delivered to air handling unit **150**.

[0096] In some embodiments, while operating hybrid adiabatic cooling system **100** in the chiller mode (dry or wet chiller mode), method **700** may include a step **765** of exchanging heat between indoor ambient air and the diverted portion of the first cooling liquid circulating from first segment **122** to second segment **124** of inner conduit loop **120**. Step **765** may include using the air handling unit **150** to exchange between indoor ambient air and the diverted portion of the first cooling liquid.

[0097] In some embodiments, method **700** may include additional steps, such as a step of determining to operate the hybrid adiabatic cooling system **100** in a dry chiller mode or a wet chiller mode based on a temperature of outdoor ambient air drawn through hybrid adiabatic cooler **130**. For example, the step may include using controller **600** to determine whether the outdoor ambient air temperature is above the second temperature threshold (82° F.) and operate in the wet chiller mode when the outdoor ambient air temperature is above the second temperature threshold.

[0098] All temperatures and fluid flow rates described herein are approximate and illustrative of one or more embodiments according to the present disclosure.

[0099] It is to be appreciated that the Detailed Description section, and not the Summary and Abstract sections, is intended to be used to interpret the claims. The Summary and Abstract sections may set forth one or more but not all exemplary embodiments of the present embodiments as contemplated by the inventor(s), and thus, are not intended to limit the present embodiments and the appended claims in any way.

[0100] The present disclosure has been described above with the aid of functional building blocks illustrating the implementation of specified functions and relationships thereof. The boundaries of these functional building blocks have been arbitrarily defined herein for the convenience of the description. Alternate boundaries can be defined so long as the specified functions and relationships thereof are appropriately performed.

[0101] The foregoing description of the specific embodiments will so fully reveal the general nature of the embodiments that others can, by applying knowledge within the skill of the art, readily modify and/or adapt for various applications such specific embodiments, without undue experimentation, without departing from the general concept of the present disclosure. Therefore, such adaptations and modifications are intended to be within the meaning and range of equivalents of the disclosed embodiments, based on the teaching and guidance presented herein. It is to be understood that the phraseology or terminology herein is for the purpose of description and not of limitation, such that the terminology or phraseology of the present specification is to be interpreted by the skilled artisan in light of the teachings and guidance.

Claims

1. A hybrid adiabatic cooling system for a data center, comprising: a first conduit loop including: a feed segment to direct a first cooling liquid from outside the data center to inside the data center, and a return segment to direct the first cooling liquid from inside the data center to outside the data center; a hybrid adiabatic cooler disposed outside of the data center, the hybrid adiabatic cooler including: an outlet fluidly coupled to the feed segment of the first conduit loop, and an inlet fluidly coupled to the return segment of the first conduit loop, wherein the hybrid adiabatic cooler is configured to draw outdoor ambient air to cool the first cooling liquid circulating from the inlet of the hybrid adiabatic cooler to the outlet of the hybrid adiabatic cooler; a direct liquid cooling unit disposed inside the data center, the direct liquid cooling unit including: an inlet fluidly coupled to the feed segment of the first conduit loop to receive the first cooling liquid exiting the hybrid adiabatic cooler, and an outlet coupled to the return segment of the first conduit loop to discharge the first cooling liquid to the return segment of the first conduit loop, wherein the direct liquid cooling unit is configured to transfer heat from a second cooling liquid to the first cooling liquid circulating from the inlet of the direct liquid cooling unit to the outlet of the direct liquid cooling

unit; a second conduit loop including: a first segment fluidly coupled to the feed segment of the first conduit loop to divert a portion of the first cooling liquid exiting the hybrid adiabatic cooler in the feed segment of the first conduit loop to the first segment, and a second segment fluidly coupled to the first segment of the second conduit loop and the return segment of the first conduit loop to direct the diverted portion of the first cooling liquid to the return segment of the first conduit loop; and an air handling unit disposed inside the data center, the air handling unit including: an inlet fluidly coupled to the first segment of the second conduit loop to receive the diverted portion of the first cooling liquid from the first segment of the second conduit loop, and an outlet fluidly coupled to the second segment of the second conduit loop to discharge the diverted portion of the first cooling liquid to the second segment of the second conduit loop, wherein the air handling unit is configured to transfer heat from indoor ambient air drawn through the air handling unit to the diverted portion of the first cooling liquid circulating from the inlet of the air handling unit to the outlet of the air handling unit.

2. The hybrid adiabatic cooling system of claim 1, further comprising: a chiller fluidly coupled to the first and second segments of the second conduit loop, wherein the chiller is configured to cool the diverted portion of first cooling liquid circulating through the first segment of the second conduit loop and heat the diverted portion of first cooling liquid circulating through the second segment of the second conduit loop.

3. The hybrid adiabatic cooling system of claim 2, wherein the first segment of the second conduit loop includes: a feed line fluidly coupled to the feed segment of the first conduit loop; a first chiller inlet line fluidly coupled to the feed line and a first inlet of the chiller; a chiller outlet line fluidly coupled to a first outlet of the chiller and the inlet of the air handling unit; and a bypass feed line fluidly coupled to the feed line and the chiller outlet line.

4. The hybrid adiabatic cooling system of claim 3, wherein the second segment of the second conduit loop includes: a discharge line fluidly coupled to the outlet of the air handling unit; a second chiller inlet line fluidly coupled to the discharge line and a second inlet of the chiller; a second chiller outlet line fluidly coupled to a second outlet of the chiller and the return segment of the first conduit loop; and a bypass outlet line fluidly coupled to the discharge line and the return segment of the first conduit loop.

5. The hybrid adiabatic cooling system of claim 4, wherein the chiller includes: an evaporator section disposed between the first inlet and the first outlet of the chiller, wherein the evaporator section is configured to exchange heat between the first cooling liquid and a working fluid such that the working fluid is evaporated to a gaseous state; and a condenser section disposed between the second inlet and the second outlet of the chiller, wherein the condenser section is configured to exchange heat between the first cooling liquid and the working fluid such that the working fluid is condensed to a liquid state.

6. The hybrid adiabatic cooling system of claim 4, further comprising: a first valve fluidly coupled to the feed line, the first chiller inlet line, and the bypass feed line of the first segment of the second conduit loop, wherein the first valve is configured to switch between a first position and a second position, wherein set in the first position, the first valve is configured to direct the first cooling liquid from the feed line to the bypass feed line, and wherein set in the second position the first valve is configured to direct the first cooling liquid from the feed line to the first chiller inlet line.

7. The hybrid adiabatic cooling system of claim 6, further comprising: a controller in communication with the first valve, wherein the controller is configured to set the first valve in the first position when a temperature of the outdoor ambient air is below a first threshold temperature and in the second position when the temperature of the outdoor ambient air is below a first threshold temperature.

8. The hybrid adiabatic cooling system of claim 7, further comprising: a second valve fluidly coupled to the discharge line, the second chiller inlet line, and the bypass outlet line of the second segment of the second conduit loop, wherein the second valve is configured to switch between a

first position and a second position, wherein set in the first position, the second valve is configured to direct the first cooling liquid from the discharge line to the bypass outlet line, and wherein set in the second position, the second valve is configured to direct the first cooling liquid from the discharge line to the second chiller inlet line.

9. The hybrid adiabatic cooling system of claim 8, wherein the controller is configured to set the second valve in the first position when the temperature of the outdoor ambient air is below the first threshold temperature and in the second position when the temperature of the outdoor ambient air is above the first threshold temperature.

10. The hybrid adiabatic cooling system of claim 7, wherein the hybrid adiabatic cooler includes: an air inlet; an air outlet; a blower configured to draw the outdoor ambient air from the air inlet to the air outlet; a tube coil disposed between the air inlet and the air outlet, wherein the tube coil is fluidly coupled to the inlet and outlet of the hybrid adiabatic cooler; and an adiabatic pad disposed at the air inlet, wherein the adiabatic pad is configured to moisturize the outdoor ambient air drawn through the air inlet, wherein the hybrid adiabatic cooler is configured to operate in a dry mode such that the adiabatic pad is dry and in a wet mode such that the adiabatic pad is wet to moisturize the outer ambient air drawn through the air inlet.

11. The hybrid adiabatic cooling system of claim 10, wherein the controller is in communication with the hybrid adiabatic cooler, and the controller is configured to set the hybrid adiabatic cooler in the dry mode when the temperature of the outdoor ambient air is below a second threshold temperature and in the wet mode when the temperature of the outdoor ambient air is above the second threshold temperature.

12. The hybrid adiabatic cooling system of claim 11, wherein the first threshold temperature is in a range from 55° F. to 75° F., and the second threshold temperature is in a range from 70° F. to 95° F.

13. The hybrid adiabatic cooling system of claim 1, wherein the first cooling liquid is water, and the second cooling liquid is a dielectric coolant.

14. The hybrid adiabatic cooling system of claim 2, further comprising: a controller configured to operate the hybrid adiabatic cooling system in a free cooling mode when a temperature of the outdoor ambient air is less than a first threshold temperature and a chiller mode when the temperature of the outdoor ambient air is greater than the first threshold temperature, wherein the hybrid adiabatic cooling system is set in the free cooling mode, the first cooling liquid in the first segment of second conduit loop bypasses the chiller, and wherein the hybrid adiabatic cooling system is set in the chiller mode, the first cooling liquid in the first segment of second conduit loop circulates through the chiller.

15. The hybrid adiabatic cooling system of claim 14, further comprising: a first pump fluidly coupled to the first conduit loop, the first pump configured to propel the first cooling liquid through the first conduit loop; a second pump fluidly coupled to the second conduit loop, the second pump configured to propel the first cooling liquid through the second conduit loop.

16. The hybrid adiabatic cooling system of claim 15, wherein the controller is in communication with the first pump and the second pump, and the controller is configured to deactivate the second pump when setting the hybrid adiabatic cooling system in the free cooling mode and activate the second pump when setting the hybrid adiabatic cooling system in the chiller mode.

17. The hybrid adiabatic cooling system of claim 14, wherein the hybrid adiabatic cooler is configured to operate in a dry mode and a wet mode, wherein set in the dry mode, the hybrid adiabatic does not moisture the outdoor ambient air drawn through the hybrid adiabatic cooler, and wherein set in the wet mode, the hybrid adiabatic cooler moisturizes outdoor ambient air drawn through the hybrid adiabatic cooler.

18. The hybrid adiabatic cooling system of claim 17, wherein the controller is in communication with the hybrid adiabatic cooler, and the controller is configured to set the hybrid adiabatic cooler in the dry mode when the temperature of the outdoor ambient air is below a second threshold temperature and in the wet mode when the temperature of the outdoor ambient air is above the

second threshold temperature, and wherein the second threshold temperature is greater than the first threshold temperature.

19-21. (canceled)

22. The hybrid adiabatic cooling system of claim 2, wherein the first cooling liquid exiting the hybrid adiabatic cooler in the feed segment of the first conduit loop is set at a first temperature, and the diverted portion of the first cooling liquid exiting the chiller in the first segment of the second conduit loop is set at a second temperature that is lower than the first temperature.
